

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6				Indicative content • smart materials have properties that change with a change in their surroundings • these changes are reversible / temporary thermochromic pigments • change colour with changing temperature • uses include thermometers, cups, baby bottles and mood rings photochromic pigments • change colour with changing light (UV) • uses include lenses in sunglasses, windows and fashion jewellery and clothing hydrogels (polymer gels) • absorb many times their weight in water / expand in water • uses include nappies, watering beads and magic snow	6			6		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
				5-6 marks Good description of the properties; uses of all three smart materials <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i> 3-4 marks Description of some of the properties; some uses <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i> 1-2 marks Properties or uses of one or two smart materials <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks <i>No attempt made or answer worthy or any credit.</i>						
				Question 6 total	6	0	0	6	0	0